AlphaFloor EFB Pilot User Guide

Universal Electronic Flight Bag for X-Plane 12

CyberPatate

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Contents

1 AlphaFloor EFB User Guide 2
1.1 Supported Aircraft 3
1.2 Operating Modes 4
1.3 Quick Navigation 4
2 Installation 4
2.1 System Requirements 4
2.2 Step-by-Step Installation 5
2.3 First Launch & Verification 6
2.4 Connecting External Devices (iPad, Android, Laptops) 6
2.5 Updating AlphaFloor EFB 6
3 Licensing & Free Trial 7
3.1 7-Day Free Trial 7
3.2 Purchasing a License 7
3.3 Activating Your License 7
3.4 Multi-Device Policy 8
3.5 Offline License Verification 8
3.6 Refund Policy 8
4 Interface & Controls 8
4.1 Screen Layout 8
4.2 Responsive Layout & Small Tablet Adaptation 9
4.3 Interaction Methods 10
5 SimBrief Operational Flight Plan (OFP) 10
5.1 Setting Up Your SimBrief Account 11
5.2 Application Sections 11
6 Takeoff Performance Calculator 16
6.1 Takeoff Calculation Workflow 17
7 Landing Performance Calculator 18
7.1 Landing Calculation Workflow 18
8 Weight & Balance 19
8.1 Operating Modes 19
8.2 Configuration Workflow 20
8.3 Interactive Seat Map 20

9 Charts & Navigation 21
9.1 Navigraph Account Linking 21
9.2 Navigating Airport Charts 21
9.3 Interactive Chart Viewer Features 23
10 Weather 25
10.1 Display Modes 25
10.2 Interactive 2D Weather Map 25
10.3 Vertical Atmospheric Profile 26
10.4 Forecast Timeline & 4D Flight Sync 26
10.5 Data Providers & Rate Limiting 27
11 Documents Viewer 27
11.1 Adding Your Own Documents 27
11.2 Document Viewer Controls 27
12 Checklists 28
12.1 Airframe-Tailored Checklist Flows 28
12.2 Operating Checklists 29
12.3 Custom Checklists (`user_checklists.json`) 30
13 Ground Services 31
13.1 Dynamic Overhead Aircraft Schematic 31
13.2 Commercial Airliner Operations 32
13.3 General Aviation Operations 33
14 Settings & System Configuration 34
14.1 1. License Management 34
14.2 2. Aircraft Unit System 34
14.3 3. Account Integrations 34
14.4 4. Web Browser Settings 35
14.5 5. Charts Mode 35
14.6 6. Remote Access (Wi-Fi LAN Streaming) 35
14.7 7. Synchronize EFBs 35
14.8 8. Virtual Keyboard 35
15 Troubleshooting & FAQ 35
15.1 Remote Tablet & Wi-Fi Streaming Issues 36
15.2 SimBrief & Charts Issues 36
15.3 Weather Issues 36
15.4 Licensing Issues 36
15.5 Contact Support 37

1 AlphaFloor EFB User Guide

Welcome to the **AlphaFloor EFB** user guide. AlphaFloor EFB is a modern Universal Electronic Flight Bag designed for X-Plane 12. It provides comprehensive operational flight planning, performance calculations, aircraft weight and balance management, charts, checklists, documents, weather radar, and ground servicing tools.

AlphaFloor EFB can be operated directly inside a 3D cockpit tablet, in a native 2D floating popup window, or streamed over your local network to any tablet, iPad, phone, or secondary computer.

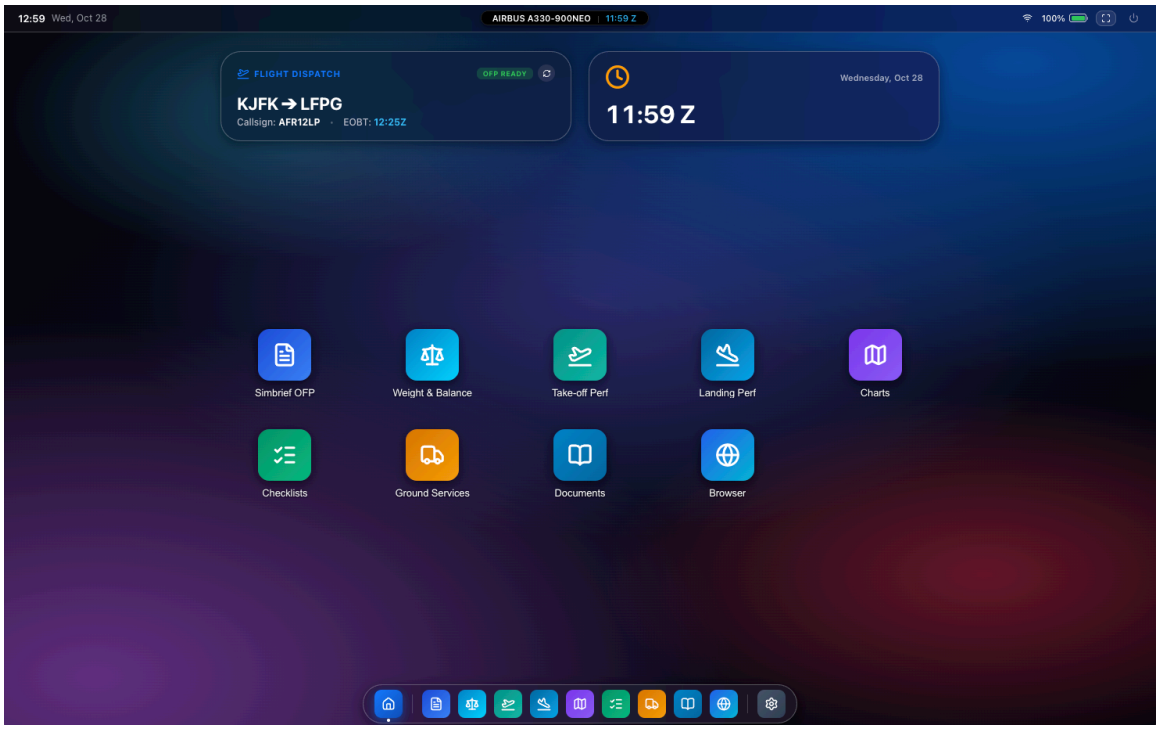


Figure 1: AlphaFloor EFB Home Screen

1.1 Supported Aircraft

AlphaFloor EFB works across **all X-Plane 12 aircraft** via its modular Aircraft Adaptation Layer, native 2D floating popup window, and remote Wi-Fi tablet streaming. In addition, dedicated airframe profiles, custom 3D cockpit integrations, and tailored operational checklists are provided for:

1.1.1 Commercial Airliners

- **ToLiss A319**
- **ToLiss A320 & A320neo**
- **ToLiss A321, A321neo, A321LR & A321XLR**
- **ToLiss A330-900**
- **ToLiss A340-600**
- **Laminar Research Airbus A330-300** (featuring native 3D tablet injection via the X-Plane Avionics API)

1.1.2 General Aviation

- **Aerobask Diamond DA42 NG** (native 3D tablet cockpit integration, tailored ground handling, and operational checklists)

- **Airfoillabs Cessna 172 Digital & Laminar Research Cessna 172** (tailored GA ground operations, tie-downs, engine covers, and full checklists)

1.1.3 Universal Fleet

- **All other X-Plane 12 Aircraft**: Operates with universal weight and fuel loading, ground operations, SimBrief OFP dispatch, Navigraph charts, live weather and radar, checklists, documents, and web browser.

1.2 Operating Modes

1. **3D Virtual Cockpit Tablet**: Renders directly onto compatible 3D cockpit tablets (`config.json` or `AviTab.json`) with native 3D mouse capture. On aircraft lacking an existing 3D tablet mesh, AlphaFloor EFB can inject a standalone 3D tablet model directly into the cockpit via the Avionics API.
2. **2D Floating Popup Window**: Open a resizable 2D floating EFB window in any aircraft via `Plugins > AlphaFloor EFB > Show EFB Window` or the `AlphaFloorEFB/toggle_window` keybind command.
3. **External Device (Wi-Fi LAN Streaming)**: Connect your iPad, Android tablet, phone, or secondary computer to the same local network as your simulation PC. Open the network address displayed in the EFB Settings to control the EFB with responsive touch controls.

1.3 Quick Navigation

- Installation Guide: How to install and verify the plugin.
- Licensing & Free Trial: Starting your 7-day free trial, purchasing a lifetime license, and multi-device activation.
- Interface & Controls: Cockpit interaction, window controls, and remote web access.
- Applications Guide: Step-by-step instructions for each built-in EFB application.
- Troubleshooting: Solutions for common setup and network issues.

2 Installation

This guide walks you through installing AlphaFloor EFB into X-Plane 12, verifying the setup, and configuring external tablet connections over your local network.

> The **AviTab** plugin should be disabled when using AlphaFloor EFB to avoid any unexpected interactions.

2.1 System Requirements

- **Flight Simulator**: X-Plane 12 (version 12.00 or later).

- **Operating System**:
  - **Windows**: Windows 10 or Windows 11 (64-bit).
  - **macOS**: macOS Monterey (12.0) or newer (Universal binary natively supporting Apple Silicon and Intel).
  - **Linux**: Ubuntu 22.04 LTS, Debian 12, Fedora 38 or equivalent 64-bit distribution with standard glibc.
- **Supported Aircraft**: Any X-Plane 12 aircraft. Dedicated airframe profiles, custom 3D integrations, and tailored operational checklists are provided for:
  - **Commercial Airliners**: ToLiss A319, A320 & A320neo, A321, A321neo, A321LR & A321XLR, A330-900, A340-600, and Laminar Research Airbus A330-300.
  - **General Aviation**: Aerobask Diamond DA42 NG, Airfoillabs Cessna 172 Digital, and Laminar Research Cessna 172.
  - **Universal Fleet**: All other X-Plane 12 aircraft via the 2D floating popup window, AviTab 3D mesh integration, and external Wi-Fi tablet streaming.

2.2 Step-by-Step Installation

2.2.1 1. Download the Release Archive

Download the latest release archive (`AlphaFloorEFB-vX.Y.Z.zip`) from the CyberPatate Website.

2.2.2 2. Extract and Move to the Plugins Folder

Extract the downloaded `.zip` file. You will see a folder named `AlphaFloorEFB`.

Move the entire `AlphaFloorEFB` folder into your X-Plane 12 plugins directory:

```
X-Plane 12/Resources/plugins/AlphaFloorEFB/
```

Or use plugin management tools such as XLauncher.

2.2.3 3. Verify Folder Structure

Confirm that the folder hierarchy looks as follows:

```
X-Plane 12/
\-- Resources/
    \-- plugins/
        \-- AlphaFloorEFB/
            |-- mac_x64/
            |   \-- AlphaFloorEFB.xpl
            |-- win_x64/
            |   \-- AlphaFloorEFB.xpl
            |-- lin_x64/
            |   \-- AlphaFloorEFB.xpl
            |-- config.json
            |-- user_checklists.json.example
            |-- skunkcrafts_updater.cfg
            \-- user-guide.pdf
```

Important: Do not rename the `AlphaFloorEFB` folder. Keep all files in their respective subdirectories.

2.3 First Launch & Verification

1. Launch **X-Plane 12**.
2. From the flight setup menu, load any aircraft in X-Plane 12.
3. Once the cockpit finishes loading, view the 3D cockpit tablet (on aircraft with 3D tablet integration) or open the 2D floating window from **Plugins > AlphaFloor EFB > Show EFB Window**.
4. The tablet screen will power on and display the AlphaFloor EFB home screen.
5. Click anywhere on the screen with your mouse to interact with the applications.

2.4 Connecting External Devices (iPad, Android, Laptops)

You can use any external tablet, iPad, smartphone, or laptop as a standalone touch EFB while flying.

1. Connect your external device to the **same local Wi-Fi network** as your flight simulator PC.
2. In the 3D cockpit EFB, tap the **Settings** icon on the bottom navigation bar.
3. Locate the **Remote Tablet Access** section. You will see an address such as:

   `http://192.168.1.42:40899`
4. Open your preferred web browser (e.g., Safari on iPad or Chrome on Android/PC) and enter that URL.
5. The EFB interface will load with full touch functionality. Any action performed on the external device is instantly synchronized with the aircraft and the cockpit screen.

Tip for iPad users: In Safari, tap the Share button and select **Add to Home Screen**. This launches the EFB in full-screen mode without browser address bars.

2.5 Updating AlphaFloor EFB

- **Via SkunkCrafts Updater**: AlphaFloor EFB includes full SkunkCrafts Updater integration. When an update is published, open your SkunkCrafts Updater client to download and apply the update. Your license file and custom checklists are automatically preserved.
- **Manual Update**: Download the new `.zip` package from the website and overwrite the contents of `X-Plane 12/Resources/plugins/AlphaFloorEFB/`. Your license file (`AlphaFloorEFB.lic` or `license.json`) will remain intact.

3 Licensing & Free Trial

AlphaFloor EFB offers a transparent, pilot-friendly licensing model. You can test every feature without restriction for seven days before deciding to purchase.

3.1 7-Day Free Trial

Every pilot is entitled to a **7-day free trial** with full access to all applications and calculations.

1. Launch X-Plane 12 with any supported Airbus aircraft.
2. If this is your first time loading the plugin, the license prompt appears on the tablet.
3. Click **Start 7-Day Free Trial**.
4. Your trial period begins immediately. An indicator at the top of the Settings screen displays your remaining trial days.
5. You do not need to provide a credit card, email address, or account registration to use the trial.

3.2 Purchasing a License

When you are ready to purchase a permanent license:

- **Price**: **11.99 €** (single one-time payment, lifetime license).
- **In-app Purchase**: Purchase the license directly from the EFB tablet within X-Plane 12.
- **Payment Processing**: Processed securely through Stripe.
- **Updates Included**: All minor and feature updates throughout the 1.x release cycle are included free of charge.

3.3 Activating Your License

After completing your purchase on the website, you will be redirected to an order confirmation screen. Your license is automatically validated in the EFB. You also have two alternative activation methods:

3.3.1 Method 1: In-App License Key Entry

1. Copy the license key displayed on your confirmation screen (e.g., `AFEFB-XXXX-XXXX-XXXX`).
2. In X-Plane 12, open the EFB tablet and tap the **Settings** icon.
3. Select the **License** section.
4. Paste or enter your key into the **License Key** field and click **Activate**.
5. The status indicator will turn green and display **Licensed (Full Version)**.

3.3.2 Method 2: Standalone License File (`AlphaFloorEFB.lic`)

1. On your order confirmation screen, click the **Download AlphaFloorEFB.lic** button.
2. Save the file directly into your plugin directory:

```
X-Plane 12/Resources/plugins/AlphaFloorEFB/AlphaFloorEFB.lic
```

3. When you load the aircraft, AlphaFloor EFB automatically detects the file and activates your license without requiring manual key entry.

3.4 Multi-Device Policy

- **Up to 3 Personal Computers**: Your purchase allows concurrent activation on up to **three (3) personal computers** (e.g., main flight sim desktop, laptop, secondary simulator rig) owned and operated by the same pilot.
- **External Tablets Do Not Consume Activations**: iPads, Android tablets, phones, and laptops connected via local Wi-Fi to display the web interface are clients of your main simulator and **do not** count toward machine activations.

3.5 Offline License Verification

Once your license is verified, AlphaFloor EFB operates completely offline on your local computer.

3.6 Refund Policy

If you encounter technical defects or hardware incompatibilities that cannot be resolved within 14 calendar days of your purchase, contact support at contact@cyberpatate.fr for a prompt, full refund.

4 Interface & Controls

4.1 Screen Layout

The tablet display is divided into three functional areas:

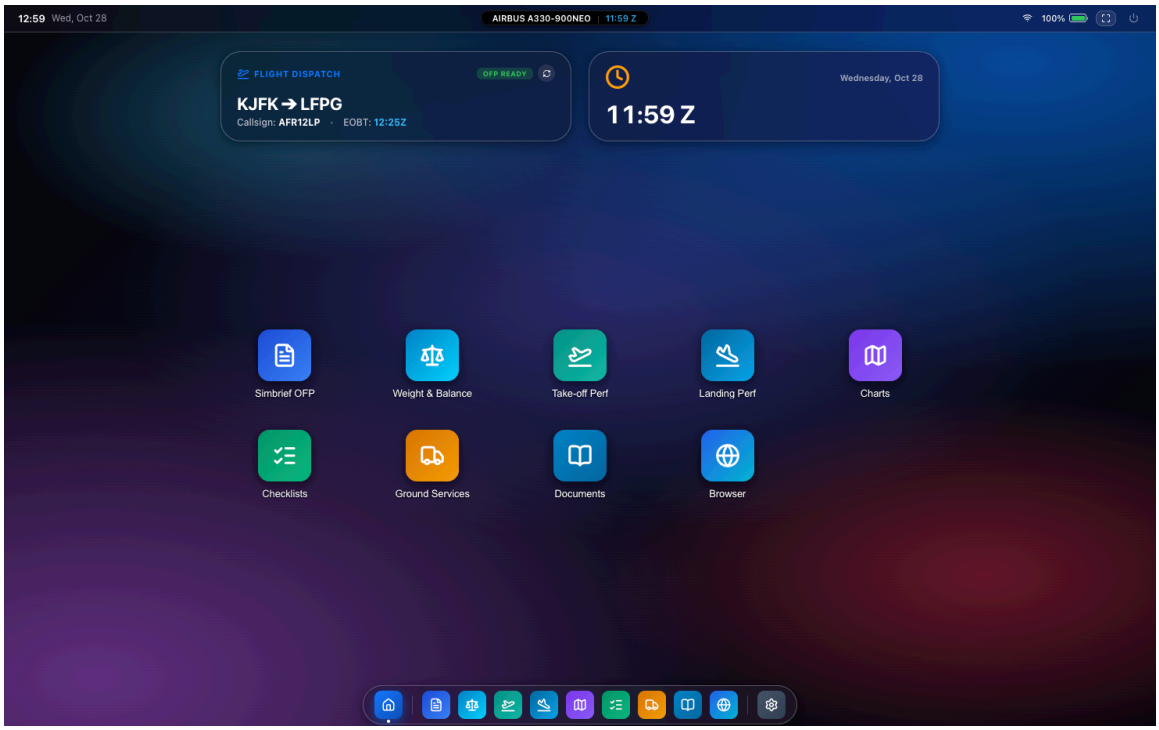


Figure 2: AlphaFloor EFB Interface & Launcher

4.1.1 1. Top Status Bar

- **Clock Display**: Displays current simulation time.
- **Aircraft Type**: Shows active aircraft variant (e.g., A319, A320neo, A321LR, A330-900, A340-600, DA42, C172).
- **Power & Indicators**: Visual indicators for battery charge level and network status.
- **Minimize Button**: Minimizes the 3D cockpit overlay on dual-mode tablet displays.

4.1.2 2. Main Content Area

This central viewport displays whichever application is currently open (e.g., Performance Calculator, SimBrief OFP, Charts, or Checklists).

4.1.3 3. Bottom Navigation Bar

- **Home Button (Left)**: Instantly returns to the application launcher grid from any screen.
- **Quick Access Icons (Middle)**: Allows switching between the main applications without returning to the home screen.
- **Settings Button (Right)**: Opens the system configuration panel.

4.2 Responsive Layout & Small Tablet Adaptation

AlphaFloor EFB automatically adapts its interface layout for different screen dimensions: * **Standard Tablets & Desktop Monitors**: Full multi-column view optimized for standard tablet aspect ratios and floating desktop windows. * **Compact / Small Displays**: Tailored compact layouts for

smaller cockpit tablet mounts, smartphones, or constrained windows, maintaining high legibility and touch target accessibility.

4.3 Interaction Methods

4.3.1 In-Cockpit 3D Interaction

- **Left Click**: Tap any button, tab, input field, or checklist item.
- **Double Left Click**: Quick zoom in/zoom out on documents and charts.
- **Mouse Drag**: Hold the left mouse button and drag to scroll through documents, charts and list items.
- **Text Input**: Click inside an input field (such as an airport ICAO code or gross weight) and type directly on your physical keyboard or use the virtual keyboard.
- **2D Floating Popup Window**: Open or close the standalone 2D floating EFB window at any time from **Plugins > AlphaFloor EFB > Show EFB Window** or by binding a key/button to `AlphaFloorEFB/toggle_window`.
- **Minimizing the 3D EFB Overlay**: On aircraft with a dual-mode 3D cockpit tablet, click the **Minimize** icon in the top status bar to temporarily hide the AlphaFloor EFB 3D overlay and access the underlying cockpit screen. To restore AlphaFloor EFB, tap the **AviTab** tab on the cockpit tablet screen, execute the `AlphaFloorEFB/toggle_overlay` command, or toggle **Enable EFB Overlay** in the *Plugins* menu.

4.3.2 External Tablet & Mobile Interaction

When viewing the EFB on an iPad, Android tablet, phone, or laptop:

- **Tap**: Equivalent to a mouse click.
- **Touch Drag**: Smooth inertial scrolling through checklists, flight logs, and weather reports.
- **Pinch to Zoom**: Zoom in and out of georeferenced airport diagrams and enroute charts.
- **Orientation Support**: The interface automatically reorganizes its layout when switching between landscape and portrait orientations.
- **Full-Screen / Standalone Mode**:
  - **iPad / iOS**: In Safari, tap the Share button and select **Add to Home Screen**. Launching AlphaFloor from the home screen icon runs it in standalone mode without Safari's address bar or navigation buttons.
  - **Android / Desktop**: Tap the **Maximize / Fullscreen** button in the top status bar (beside the battery indicator) or select **Add to Home screen** in your browser menu.

5 SimBrief Operational Flight Plan (OFP)

The **SimBrief OFP** application integrates with your SimBrief account to display your full operational flight plan, fuel calculations, route waypoints, and weather reports directly in the cockpit.

5.1 Setting Up Your SimBrief Account

1. Open the **SimBrief** app from the EFB home screen (or navigate to **Settings**).
2. Enter your **SimBrief Username** or numeric **Pilot ID**.
3. Tap **Save & Fetch**.
4. AlphaFloor EFB will query the SimBrief API and load your latest dispatch package. Your username is saved locally, so you only need to enter it once.

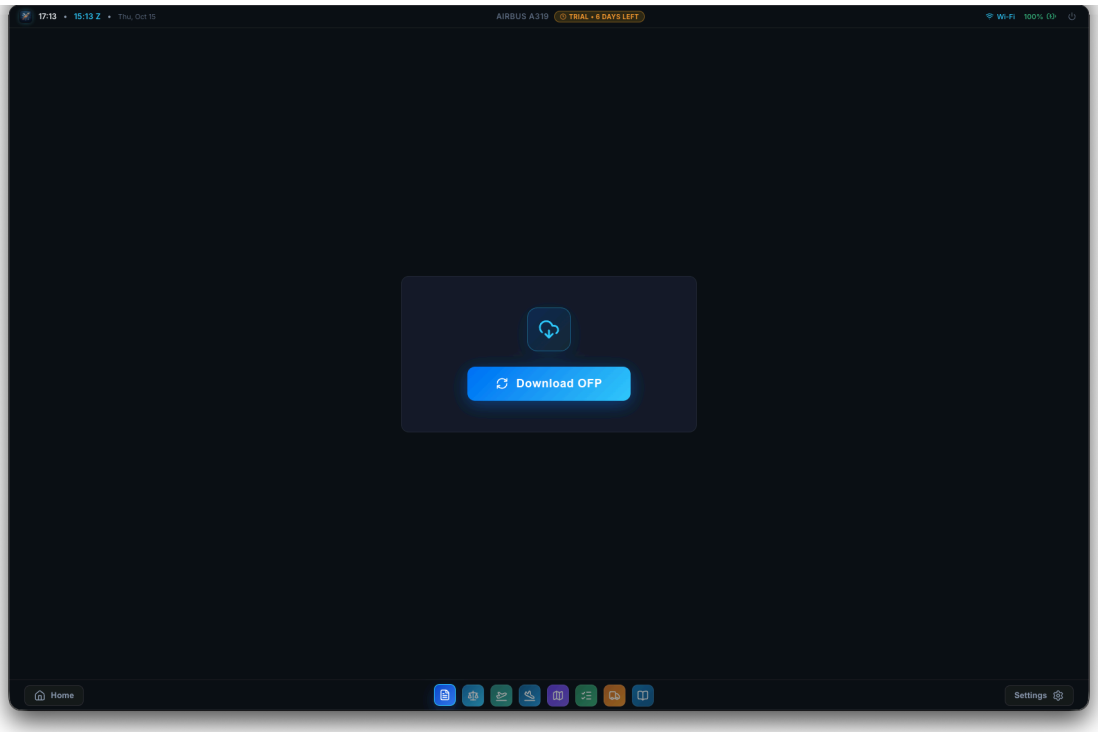


Figure 3: SimBrief Download OFP

5.2 Application Sections

5.2.1 1. Flight Overview

Displays critical flight details at the top of the screen:

- **Flight Number & Call sign** (e.g., `AFR1234`)
- **Departure & Arrival Airports** with scheduled departure times and estimated flight duration
- **Planned Runways & ATC Filed Route**: Filed route text includes planned departure and arrival runways for streamlined FMC/MCDU flight plan entry.
- **Alternate Airport(s)** with distance and reserve fuel requirements
- **Cruising Altitude / Initial Flight Level** (e.g., `FL350`)
- **Airframe Type & Registration**

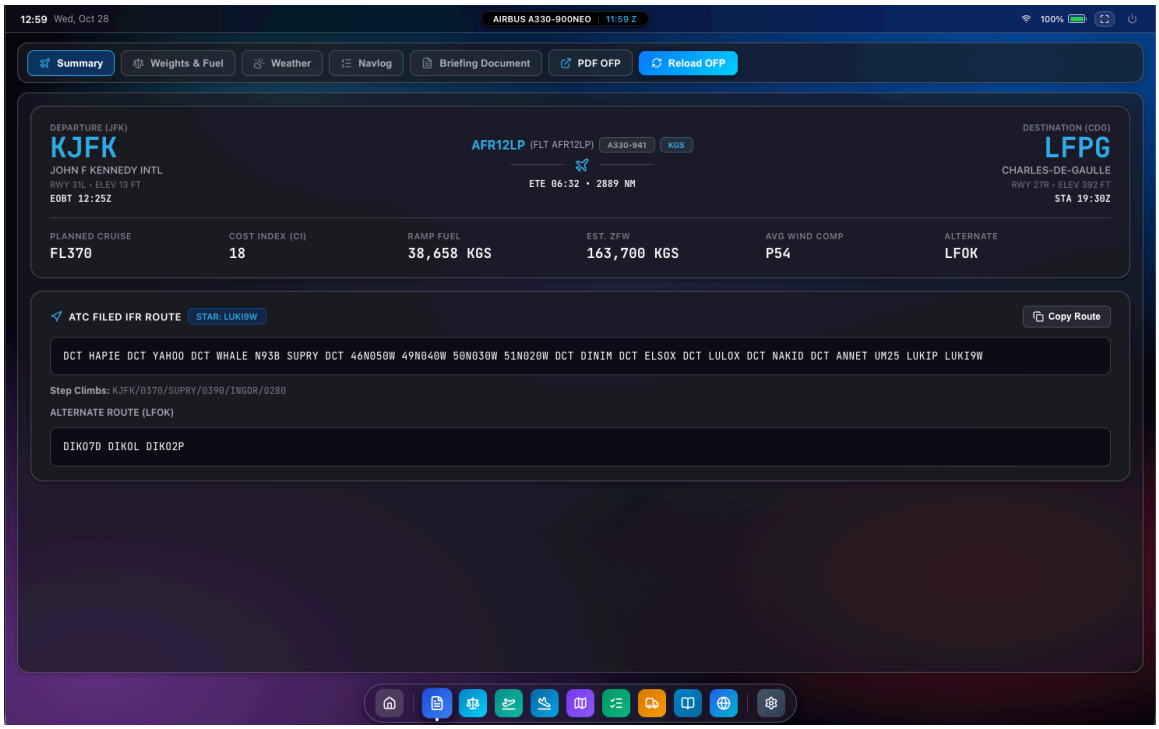


Figure 4: SimBrief OFP Summary

5.2.2 2. Fuel & Weight Breakdown

Summarizes planned fuel figures and weights for easy cross-checking with your MCDU. The app automatically detects and displays all quantities in either **Kilograms (KGS)** or **Pounds (LBS)** according to the units selected in your dispatch flight plan:

- **Block Fuel**: Total fuel required on board before pushback.
- **Trip Fuel**: Calculated burn for the planned route.
- **Contingency & Reserve Fuel**: FAR/EASA regulatory reserves.
- **Alternate Fuel**: Fuel required to divert to the planned alternate.
- **Zero Fuel Weight (ZFW)**: Planned aircraft empty weight plus payload.
- **Takeoff Weight (TOW)**: Calculated weight at brake release.

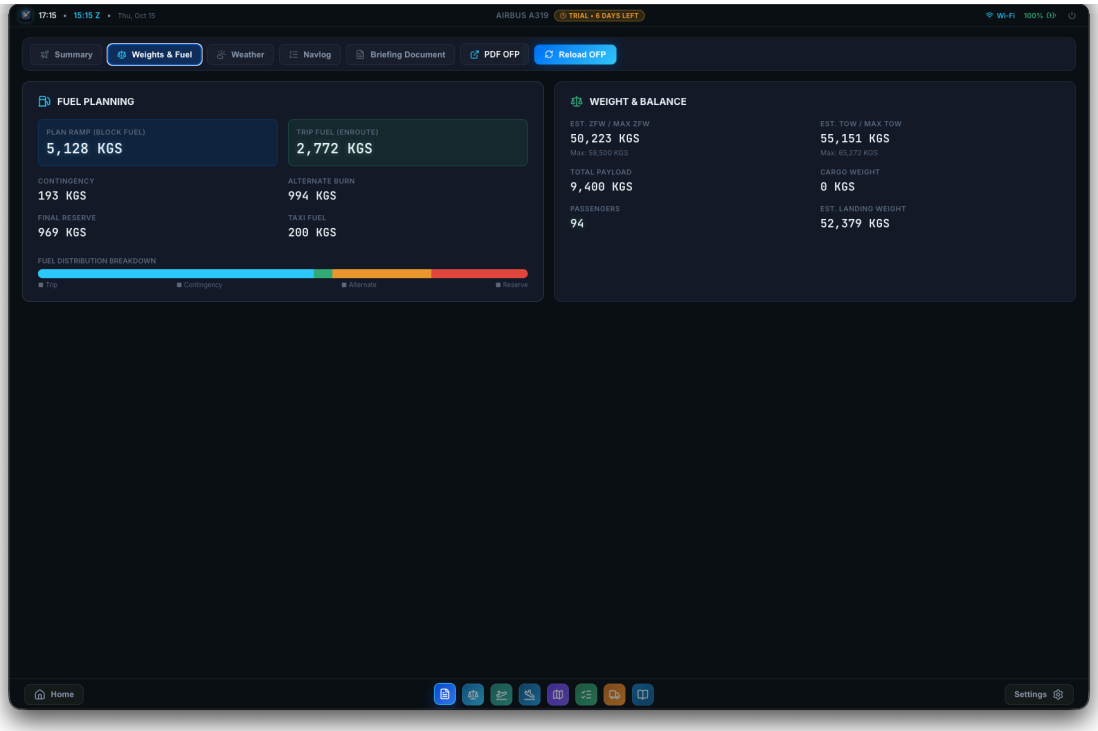


Figure 5: SimBrief Fuel & Weights

5.2.3 3. Route Navigation Log (Navlog)

Provides a scrollable navigation log listing:

- Every airway, fix, and waypoint along the cleared routing.
- Planned flight levels.
- Forecast winds aloft (direction and speed).
- Leg distance and estimated time.
- Estimated Fuel On-board and editable Actual Fuel On-board entries to track fuel burn during flight.

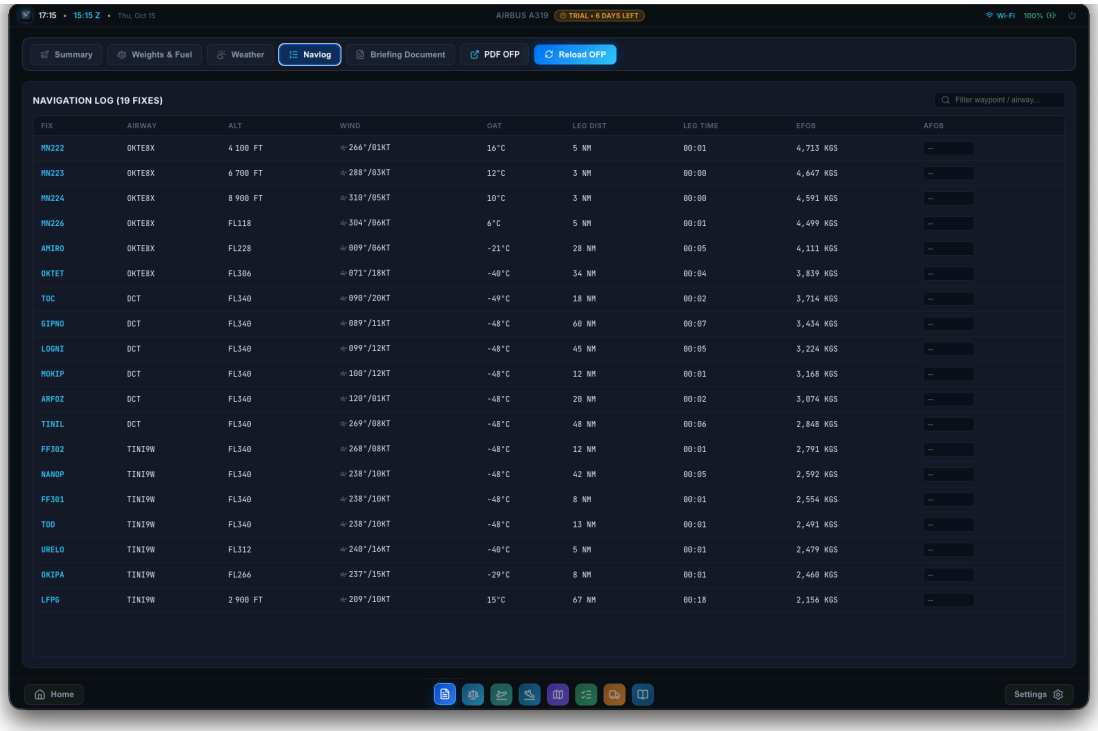


Figure 6: SimBrief Navigation Log

5.2.4 4. Weather, METAR & TAF

Aggregates real-time weather reports for all airports along your itinerary:

- **Departure Weather**: Raw METAR, decoded wind, altimeter setting (QNH/inHg), temperature, and runway surface reports.
- **Destination Weather**: Current conditions and terminal aerodrome forecasts (TAF).
- **Enroute Alternates**: Quick-reference weather cards for emergency diversion planning.

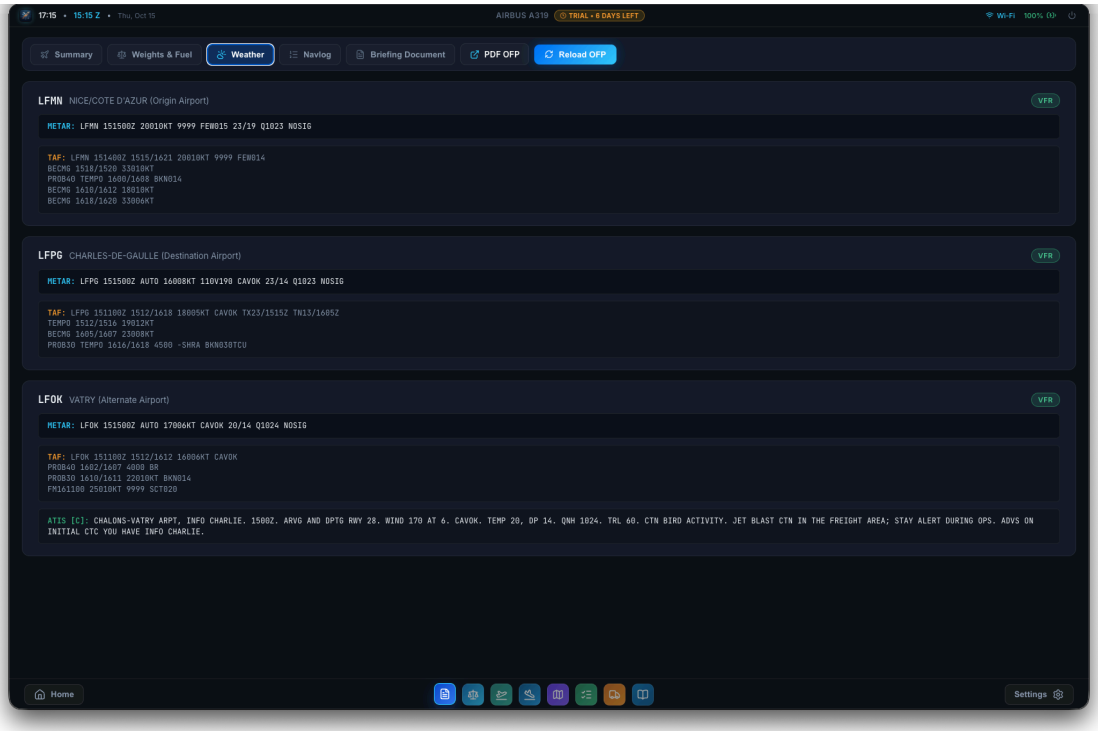


Figure 7: SimBrief Weather & METAR

5.2.5 5. Full PDF Briefing Package

View the complete, multi-page operational flight plan document formatted exactly as generated by your airline dispatch:

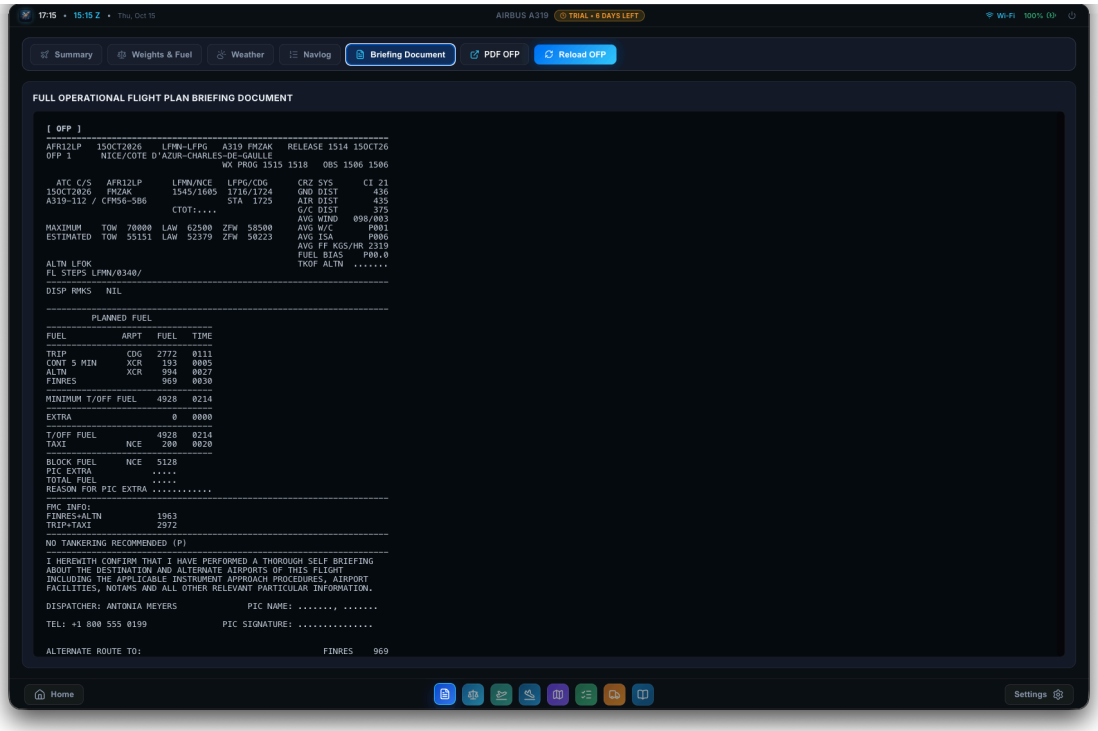


Figure 8: SimBrief OFP PDF

6 Takeoff Performance Calculator

The **Takeoff Performance** application computes precise departure performance figures, optimal thrust settings, and reference takeoff speeds.

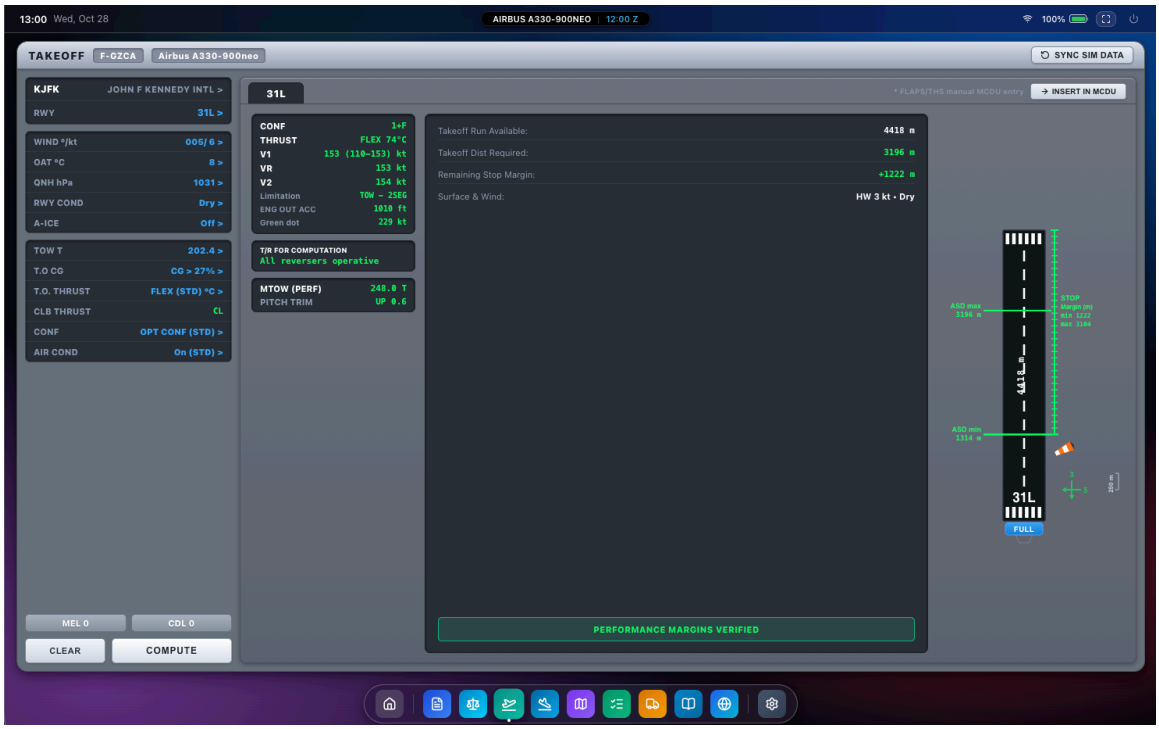


Figure 9: Takeoff Performance Calculator

6.1 Takeoff Calculation Workflow

1. **Airport & Runway Selection**: Enter the 4-letter ICAO code for your departure aerodrome. Select the active runway or designated intersection departure. Available runway lengths (TORA, TODA, ASDA) and elevation update automatically.
2. **Atmospheric Conditions**: Verify surface wind direction/speed, outside air temperature (OAT), and barometric altimeter setting (in hPa or inHg).
3. **Aircraft Configuration**: Confirm takeoff gross weight (in Tonnes or 1000 lbs), air conditioning packs status (ON/OFF), anti-ice configuration (ENG or ENG+WING), and desired flap setting.
4. **Compute Performance**: Tap **Compute** to execute the calculation.
5. **Review Results**:
   - **Assumed / FLEX Temperature**: Maximum permissible thrust reduction temperature.
   - **V-Speeds**: Decision speed (V_1), Rotation speed (V_R), and Engine-out safety speed (V_2).
   - **Trim & Speeds**: Recommended pitch trim (THS) and clean climb speed (Green Dot).
   - **Runway Margins**: Required takeoff roll distance and stop margin remaining.
6. **MCDU / FMGS Insertion**: Tap **Insert in MCDU** to transfer computed V-speeds and FLEX temperature directly into the aircraft FMGS scratchpad on supported aircraft.

[!NOTE] Due to aircraft FMGS dataref limitations, the takeoff flap and pitch trim setting (FLAPS/THS, e.g. `1/UP0.5`) cannot be inserted automatically via datarefs. Enter this figure manually into line select key 3R on the MCDU PERF TAKEOFF page.

7 Landing Performance Calculator

The **Landing Performance** application calculates approach reference speeds, stopping distances, factoring margins, and engine-out go-around climb gradients for arrival planning.

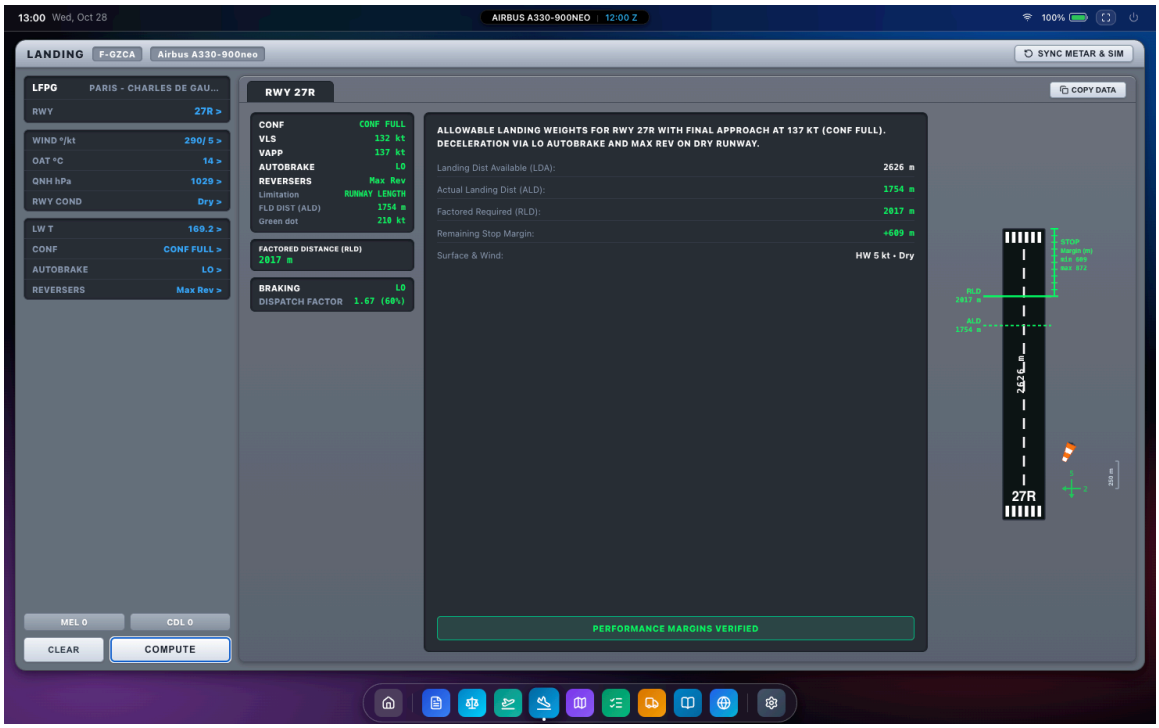


Figure 10: Landing Performance Calculator

7.1 Landing Calculation Workflow

Prior to descent (during the descent and arrival briefing):

1. **Airport & Runway Selection**: Select your planned destination or diversion aerodrome and landing runway. Runway length, slope, and elevation are loaded automatically.
2. **Weather & Runway Surface Conditions**: Enter surface wind, temperature, barometric pressure (in hPa or inHg), and runway condition (Dry, Good, Medium, Poor).
3. **Aircraft Landing Weight**: Confirm estimated landing weight (in Tonnes or 1000 lbs according to your configured unit system in Settings). When syncing with SimBrief, planned landing weights from LBS or KG flight plans are imported seamlessly.
4. **Flaps & Deceleration Devices**: Select landing flap setting, reverse thrust mode, and wheel braking method (Autobrake Low/Medium/Max or Manual).
5. **Compute Performance**: Tap **Compute** to evaluate arrival performance.

6. **Review Results**:
   - **Approach Speed (V_{APP})**: Target approach speed including wind correction.
   - **Landing Distances**: Operational landing distance (unfactored) and dispatch factored landing distance.
   - **Runway Margin**: Visual stop margin indicator showing remaining runway beyond rollout.
   - **Engine-Out Go-Around Gradient**: Computes the single-engine missed approach climb gradient percentage to verify compliance with published missed approach procedure gradient requirements.

8 Weight & Balance

The **Weight & Balance** application manages aircraft passenger distribution, cargo hold loading, fuel uplift, and Center of Gravity (CG) envelope calculations.

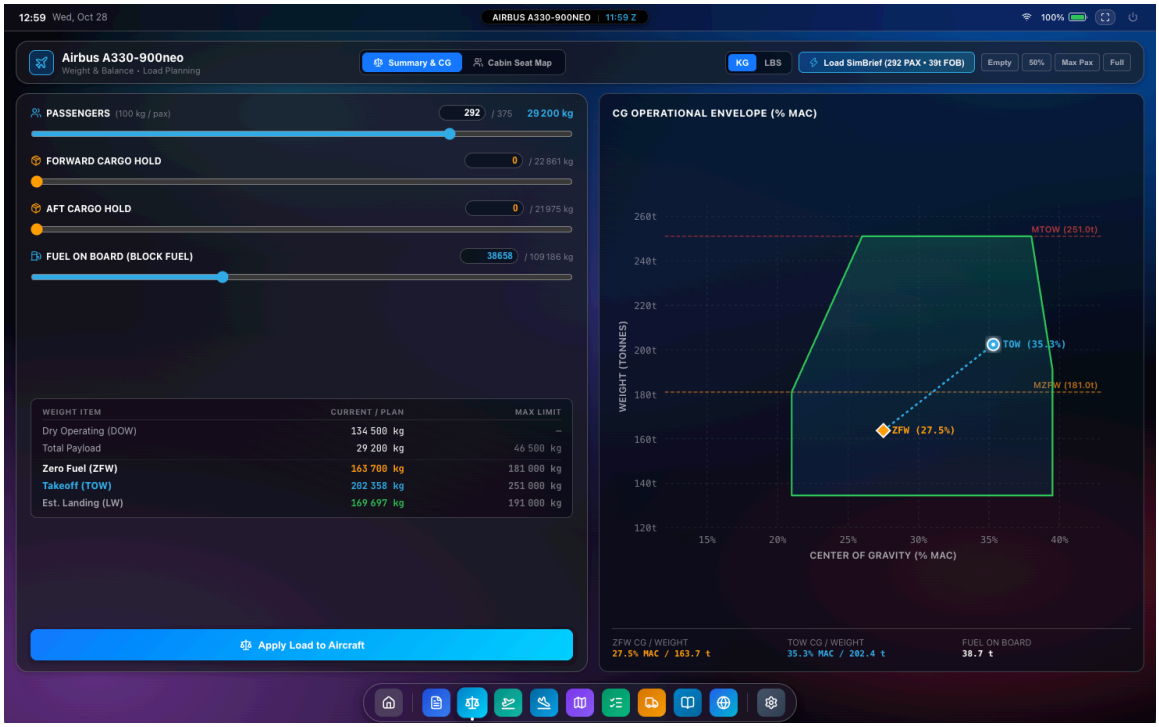


Figure 11: Weight & Balance

8.1 Operating Modes

8.1.1 1. Extended Airframe Profiles

For aircraft with dedicated airframe profiles (including Airbus A319, A320, A321, A330, and A340): * **Interactive Cabin Seat Map**: Board passengers by cabin zones or individually using the visual seat map. * **Cargo Hold Distribution**: Distribute freight across forward, aft, and bulk compartments. * **Dynamic CG Flight Envelope**: Real-time Center of Gravity (% MAC)

calculations plotted against certified forward and aft envelope limits for Zero Fuel Weight (ZFW), Takeoff Weight (TOW), and Landing Weight (LAW). * **THS Pitch Trim Calculation**: Computes the exact stabilizer pitch trim setting (e.g., `UP 1.2` or `DN 0.5`) based on computed TOW CG.

8.1.2 2. Universal Aircraft Mode

For all other X-Plane 12 aircraft: * Utilizes standard X-Plane payload stations and fuel tanks. * Allows reviewing total payload, passenger weight, cargo, and fuel distribution. * Enables one-click payload and fuel loading from SimBrief dispatch plans.

8.2 Configuration Workflow

1. **Unit System Selection**: Select your preferred unit system (**KG** or **LBS**) using the header toggle. The app automatically defaults to the system specified in your active SimBrief OFP, or the default configured in Settings.
2. **Passenger & Cargo Loading**: Adjust passenger count, cabin distribution, and cargo weights. When clicking **Load SimBrief**, planned fuel and cargo figures are imported and converted automatically.
3. **Fuel Uplift**: Set the desired fuel quantity across available fuel tanks.
4. **Envelope Verification**: Verify that the calculated Zero Fuel Weight CG and Takeoff Weight CG (% MAC) remain securely within certified operational limits.
5. **Apply to Aircraft**: Tap **Apply to Aircraft** to synchronize fuel and payload weights directly into the simulator.

8.3 Interactive Seat Map

For supported aircraft, tap the seat map toggle to inspect and adjust passenger seating arrangements dynamically:

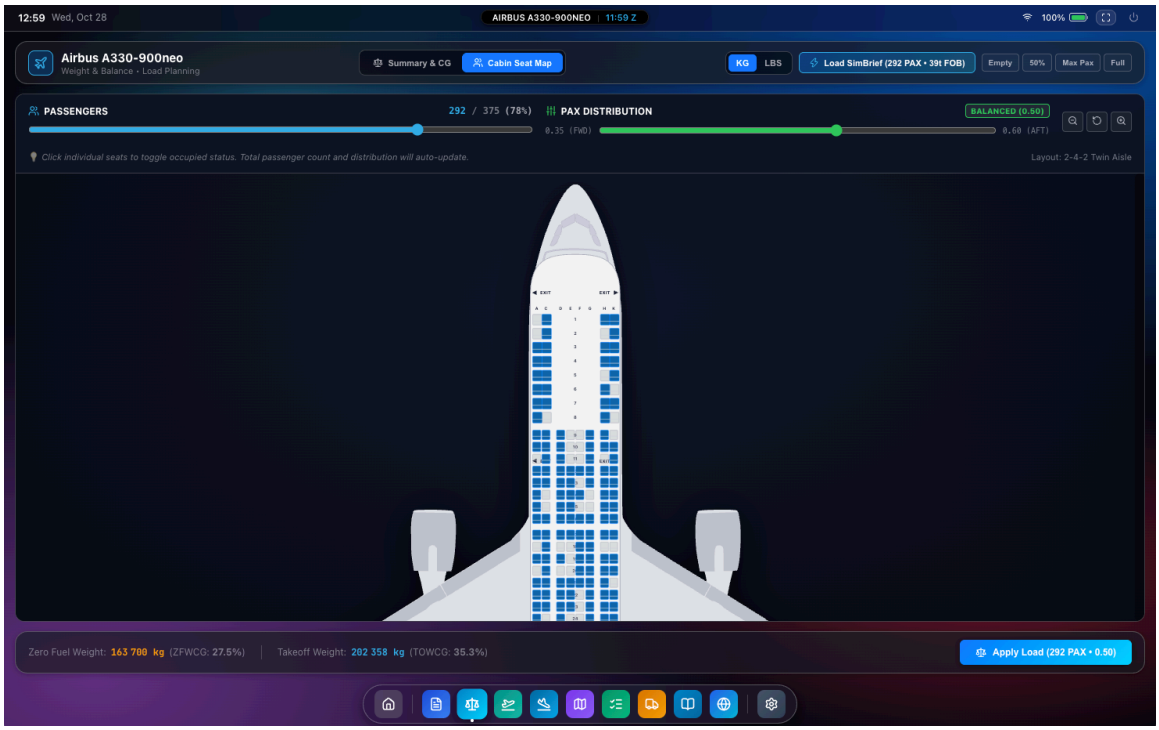


Figure 12: Interactive Seatmap

9 Charts & Navigation

The **Charts** application displays aerodrome diagrams, departure routes (SID), terminal arrivals (STAR), and approach plates directly on the tablet screen with real-time aircraft position overlay.

Important Note: A Navigraph subscription is required for this application to work.

9.1 Navigraph Account Linking

AlphaFloor EFB integrates directly with Navigraph Charts:

1. Open the **Charts** or the **Settings** application on the EFB.
2. If not previously signed in, tap **Log in to Navigraph**.
3. A Navigraph login page is displayed. You can log in with your Navigraph credentials.
4. The tablet will automatically refresh and connect to the worldwide Jeppesen chart database.

9.2 Navigating Airport Charts

9.2.1 1. Pinned Airports

When a SimBrief flight plan is active, AlphaFloor EFB automatically pins your:

- **Departure Airport**

- **Destination Airport**
- **Enroute Alternate(s)**

Tap any pinned airport badge at the top of the screen to view its chart directory.

9.2.2 2. Manual Airport Search

To search for any global airport:

1. Tap the search bar in the top navigation header.
2. Type the 4-letter ICAO code (e.g., `EGLL`, `EDDF`, `KSFO`).
3. Press Enter to view all available charts.

9.2.3 3. Chart Categories

Charts are organized into standard operational tabs:

- **APT**: Airport diagrams, parking stand charts, low-visibility taxi routes, and hot spot advisories.
- **DEP**: Standard Instrument Departures.
- **ARR**: Standard Terminal Arrival Routes.
- **APP**: Instrument Approach Procedures (ILS, RNP, RNAV, VOR, Visual).
- **REF**: Noise abatement, special departure procedures, and airport general notes.



Figure 13: Charts Overview & Directory

9.3 Interactive Chart Viewer Features

- **Moving Map Position Overlay**: On georeferenced taxi diagrams and approach plates, a magenta aircraft symbol indicates your exact real-time position and heading.
- **Day & Night Themes**: Toggle between high-contrast daylight mode and dim night mode to preserve night cockpit vision.
- **Zoom & Pan Controls**: Use your mouse click-drag in the cockpit, or natural pinch-to-zoom gestures on external tablets.
- **Rotation**: Rotate landscape and portrait charts with the 90-degree rotate button.

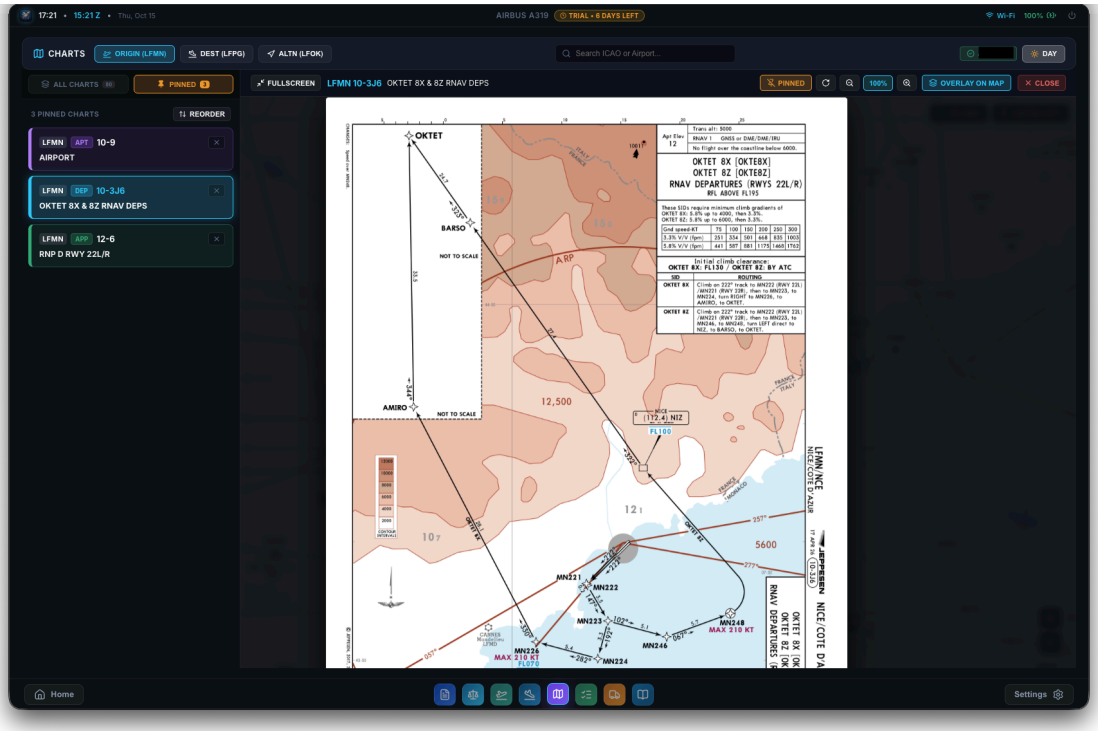


Figure 14: Interactive Chart Viewer

9.3.1 Moving Map & Georeferenced Enroute Overlay

AlphaFloor EFB overlays your planned flight plan route and real-time simulator GPS telemetry onto a fluid, interactive navigation map:

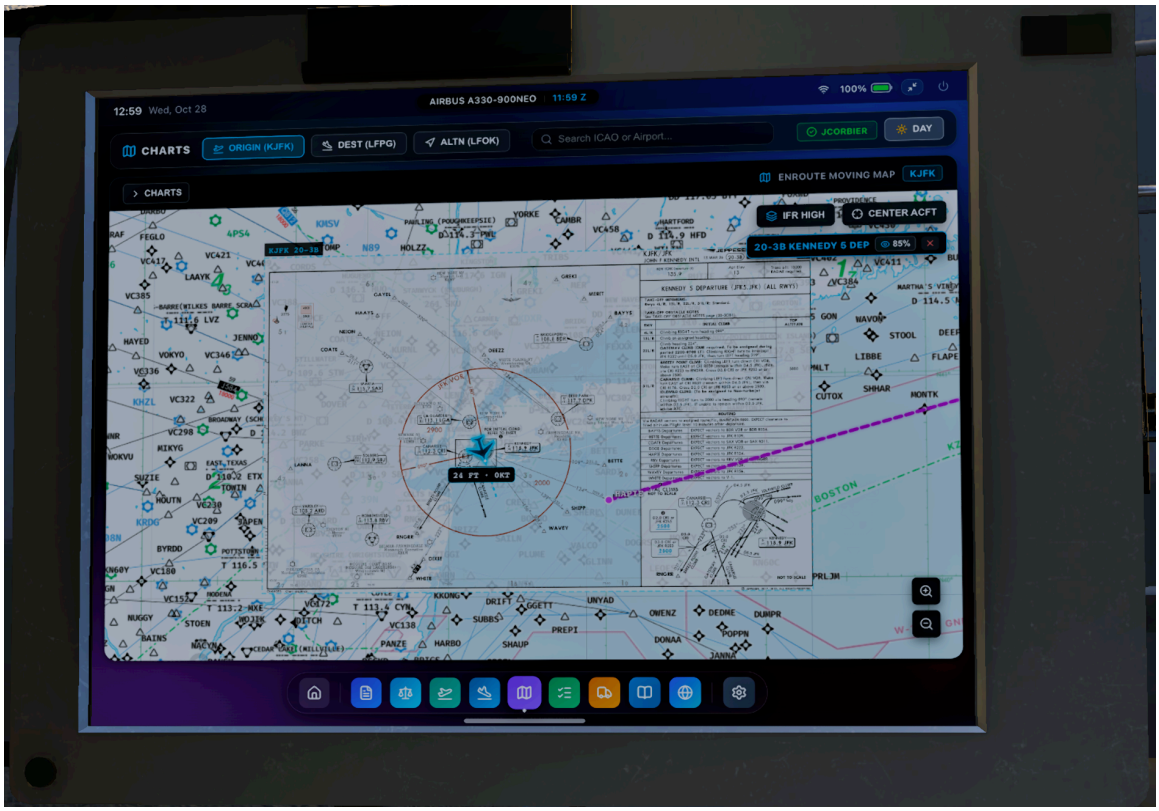


Figure 15: Chart Overlay & Enroute Moving Map

10 Weather

The **Weather** application provides enroute radar imagery and vertical atmospheric cross-sections along your planned flight route.

10.1 Display Modes

The view selector located in the top-right header allows switching between three viewing layouts:

- **Split**: Displays both the 2D interactive route map and the vertical atmospheric profile.
- **Map**: Full-screen 2D navigation map with precipitation radar, flight plan routing, waypoints, and live aircraft position.
- **Profile**: Full-screen vertical cross-section displaying cloud layers.

10.2 Interactive 2D Weather Map

The map displays your cleared routing alongside live radar:

- **Flight Route**: Depicts departure airport, intermediate waypoints, destination airport, and planned alternates.

- **Aircraft Position**: Real-time magenta aircraft symbol indicating your current GPS coordinates and true heading.
- **Precipitation Radar**: Displays radar reflectivity across light, moderate, heavy, and severe convective precipitation bands.

10.3 Vertical Atmospheric Profile

The profile view plots a longitudinal cross-section of your flight from takeoff to landing, mapped against pressure altitude (Surface up to FL450):

10.3.1 1. Flight Trajectory & Live Position

- **Planned Flight Levels**: Depicts the step-climb profile calculated in your SimBrief OFP.
- **Live Aircraft Badge**: A cyan aircraft symbol tracks your actual barometric altitude and along-track distance in real time.

10.3.2 2. Meteorological Layers & Hazards

- **Cloud Layers**: Visualizes cloud coverage with base and top altitudes, shaded according to cloud coverage percentage.
- **Freezing Level (0°C Isotherm)**: Marked as a dashed amber line indicating the zero-degree boundary throughout the route.
- **Icing**: Warning indicators highlight flight segments where supercooled moisture and freezing temperatures present icing hazards.

10.3.3 3. Interactive Atmospheric Sounding Overlay

Scrubbing your cursor or tapping anywhere along the route reveals a detailed sounding card for that specific point:

- **Distance**: Cumulative nautical miles from departure and remaining distance to destination.
- **Nearest Waypoint & Airway**: Closest navigation fix along the cleared route.
- **Planned Altitude**: Scheduled flight level from the flight plan.
- **Point Cloud Coverage**: Indicates whether the cursor is in clear air or inside a cloud deck (`In Cloud` with local coverage percentage).
- **Cloud Layers Breakdown**: Vertical listing of all cloud decks present at that waypoint, including layer index, base altitude, top altitude, and coverage percentage.

10.4 Forecast Timeline & 4D Flight Sync

The forecast toolbar at the bottom of the screen allows evaluating weather evolution over time:

- **4D Flight Sync**: Calculates atmospheric soundings dynamically matched to your aircraft's estimated time of arrival (ETA) at each waypoint along the route, taking into account departure time and elapsed flight progress.
- **Now**: Instantaneous snapshot of current weather conditions across the entire route.

- **+3H / +6H / +12H**: Forward-looking synoptic forecast projections, useful for assessing front movements and terminal weather developments for long-haul flights or delayed departures.

10.5 Data Providers & Rate Limiting

- **Precipitation Radar**: Global composite reflectivity tiles provided by RainViewer.
- **Atmospheric Soundings**: The profile view relies on the Open-Meteo API for numerical weather prediction soundings along the flight route.

Weather data is automatically cached per flight to minimize external network requests. Open-Meteo imposes request rate limits; if request limits are reached or the service is temporarily unavailable, a "Weather data temporarily unavailable" overlay is displayed until the rate-limit window resets or connectivity is restored.

11 Documents Viewer

The **Documents** application allows flight crews to view PDF flight documentation, Quick Reference Handbooks (QRH), airline Standard Operating Procedures (SOPs), and Flight Crew Operating Manuals (FCOM) directly on the tablet screen.

11.1 Adding Your Own Documents

To make custom PDF documents available in the EFB:

1. Navigate to one of the following folders:
   - Your AlphaFloor EFB plugin folder: `X-Plane 12/Resources/plugins/AlphaFloorEFB/documents/`
   - Per-aircraft folder: `X-Plane 12/Aircraft/[aircraft_folder]/efb_docs/`
   - Create the folders if they don't exist
2. Copy any standard `.pdf` files into the directory (e.g., `A320_QRH.pdf`, `Company_SOP.pdf`, `Route_Briefing.pdf`).
3. Open or refresh the **Documents** application on the EFB. Your files will appear in the document library.

11.2 Document Viewer Controls

- **Document Library**: Lists all available PDF, allowing to pin favorites for quick access.
- **Page Navigation**:
  - Tap the **Next Page** and **Previous Page** arrows to flip pages.
  - Use the **Page Jump** box to type a specific page number and press Enter.
- **Zoom & Pan**: Adjust reading scale using the Zoom In/Out controls. Pan using the mouse grab.

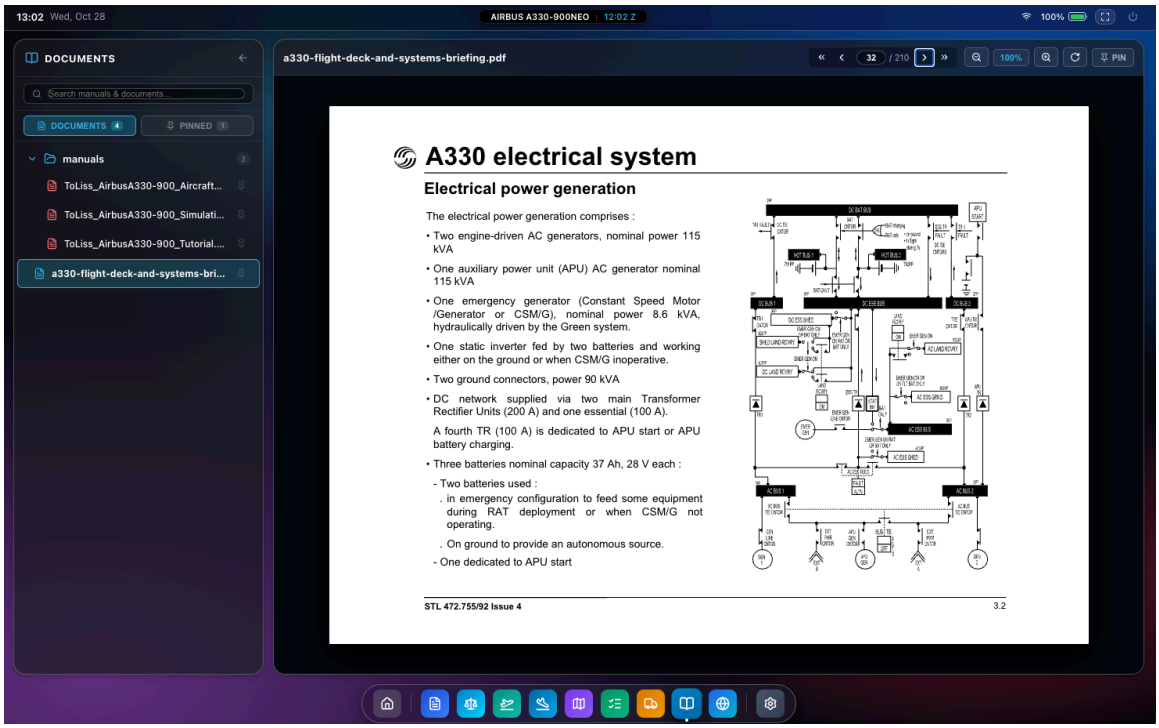


Figure 16: Documents Viewer

12 Checklists

The **Checklists** application provides interactive operational checklists designed to streamline cockpit workflows, reinforce SOP adherence, and enhance situational awareness. Checklists are automatically loaded to match the active aircraft type.

12.1 Airframe-Tailored Checklist Flows

AlphaFloor EFB includes built-in operational checklist flows tailored to specific aircraft families:

12.1.1 1. Commercial Airliner Flow

1. **Cockpit Preparation**
2. **Before Start**
3. **After Start**
4. **Taxi**
5. **Line Up**
6. **Approach**
7. **Landing**
8. **After Landing**
9. **Parking**
10. **Securing the Aircraft**

12.1.2 2. Diamond DA42 Flow

1. **Before Start**
2. **Engine Start**
3. **After Start**
4. **Taxi**
5. **Before Take-Off**
6. **After Take-Off**
7. **Descent & Approach**
8. **Landing & After Landing**

12.1.3 3. Cessna 172 Flow

1. **Preflight**
2. **Engine Start**
3. **Before Taxi**
4. **Taxi**
5. **Engine Runup**
6. **Before Takeoff**
7. **Takeoff**
8. **Climb**
9. **Cruise**
10. **Descent**
11. **Before Landing**
12. **Final Landing**
13. **After Landing**
14. **Engine Shutdown**
15. **Securing Aircraft**

12.2 Operating Checklists

- **Checking Items**: Click or tap any checklist line item. The item highlights in green with a checkmark.
- **Sub-Items**: Complex procedural checks include expandable sub-items that can be verified individually.
- **Headers & Reminders**: Clear section dividers and prominent reminder notes highlight operational limitations.
- **Phase Navigation**: Use the sidebar to quickly navigate between flight phases as your flight progresses.
- **Resetting Checklists**: Click **Reset Phase** to clear items in the active phase, or **Reset All** to reset the entire checklist suite for a new flight leg.

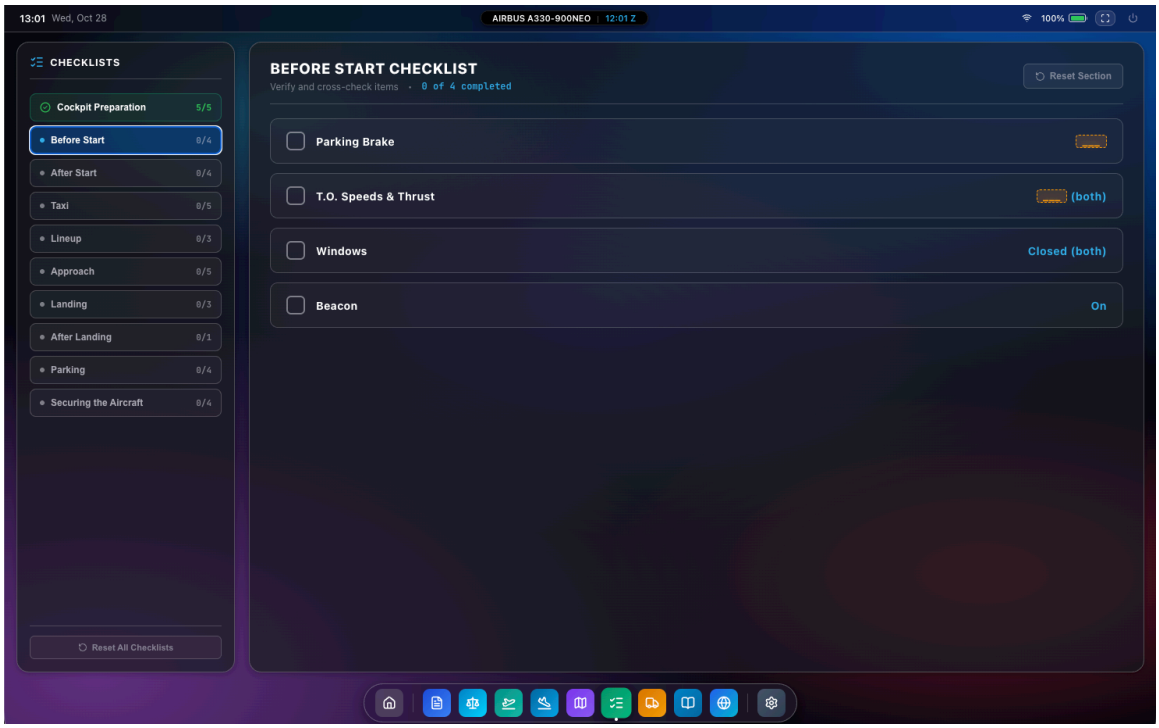


Figure 17: Interactive Checklists

12.3 Custom Checklists (`user_checklists.json`)

You can customize checklists to match your airline's specific Standard Operating Procedures (SOPs) or virtual flight operations:

1. Locate the example template in your plugin directory:

   `X-Plane 12/Resources/plugins/AlphaFloorEFB/user_checklists.json.example`

2. Make a copy of this file and rename it to:

   `user_checklists.json`

3. Open `user_checklists.json` in any text editor. It supports section titles, challenge/response pairs, sub-items, section headers, and reminder notes:

```
[
  {
    "title": "Cockpit Preparation",
    "items": [
      { "item": "Gear Pins & Covers", "check": "REMOVED" },
      { "item": "Fuel Quantity", "check": "___ KG/LB" },
      { "item": "Signs", "check": "ON" }
    ]
  },
```

```
  {
    "title": "Before Take-Off",
    "items": [
      { "item": "LIMITATION", "check": "100% = 5 MIN MAXIMUM", "reminder":
true },
      {
        "item": "ECAM Memo",
        "check": "T.O. No Blue",
        "subitems": [
          "Auto Brk MAX",
          "Signs On",
          "Flaps Conf 1+F",
          "Spoilers Armed"
        ]
      }
    ]
  }
]
```

13 Ground Services

The **Ground Services** application allows pilots to manage aircraft doors, ground servicing equipment, and external electrical power directly from the cockpit tablet. The interface dynamically adapts its schematic, door controls, and available equipment to the loaded aircraft.

13.1 Dynamic Overhead Aircraft Schematic

The application presents an interactive overhead schematic matching the geometry of the active aircraft:

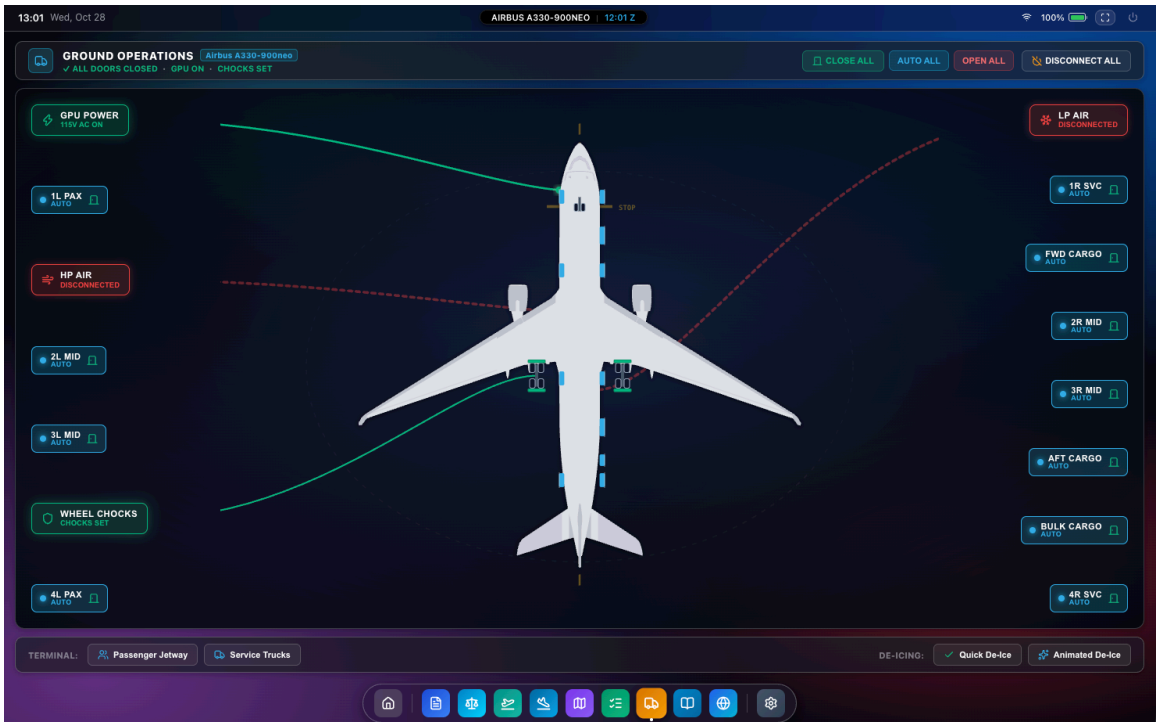


Figure 18: Aircraft Controls & Ground Services

13.2 Commercial Airliner Operations

13.2.1 Passenger & Service Doors

- **Door 1L**: Main passenger boarding door (connected to jetway or boarding stairs).
- **Door 1R**: Forward galley and catering service door.
- **Door 2L / 2R & 3L / 3R**: Overwing emergency exits and mid-cabin doors (on extended airframes).
- **Door 4L / 4R**: Aft passenger and service doors.

13.2.2 Cargo Compartments

- **Forward Cargo Door**: Opens the forward lower hold for luggage containers and pallets.
- **Aft Cargo Door**: Opens the aft lower hold.
- **Bulk Cargo Door**: Opens the rear bulk compartment.

13.2.3 Quick Actions & Automated Sequencing

- **All Doors Open**: Opens all passenger and cargo doors simultaneously for rapid turnaround.
- **All Doors Closed**: Closes and arms all doors prior to pushback.
- **Auto Doors**: On supported airliners, cycles door modes through automatic sequencing for departure and arrival.

13.2.4 Airliner Ground Equipment

- **Wheel Chocks**: Place or remove main gear chocks. (Ensure parking brake is set before removing chocks).
- **Ground Power Unit (GPU)**: Connects the ground electrical power cart.
- **HP Air**: Connects the high-pressure pneumatic air cart for engine start.
- **LP Air**: Connects the low-pressure air conditioning cart for cabin pre-conditioning.
- **Passenger Stairs**: Summons forward and aft boarding stairs for non-jetway stands.
- **Service Trucks**: Drives catering scissor-lift trucks to galley service doors.

13.3 General Aviation Operations

When operating general aviation aircraft (such as the Diamond DA42 or Cessna 172), the application switches to tailored GA ground equipment and cabin controls:

13.3.1 Cabin & Canopy Access

- **Canopy / Cabin Doors**: Open or close the pilot, passenger, and baggage doors.

13.3.2 General Aviation Servicing Equipment

- **Wheel Chocks**: Secure the aircraft wheels while parked.
- **Tie-Downs**: Secure the wing and tail tie-down ropes for ramp parking.
- **Pitot & Static Covers**: Install or remove protective pitot tube covers and static port flags.
- **Gust Locks**: Engage or release physical control surface gust locks.
- **Engine Covers**: Place protective thermal or debris cowl plugs over engine intakes.
- **Engine Pre-Heater**: Activate ground engine pre-heating for cold weather operations.
- **Fuel Grounding Cable**: Connect or disconnect the static grounding cable prior to refueling.

14 Settings & System Configuration

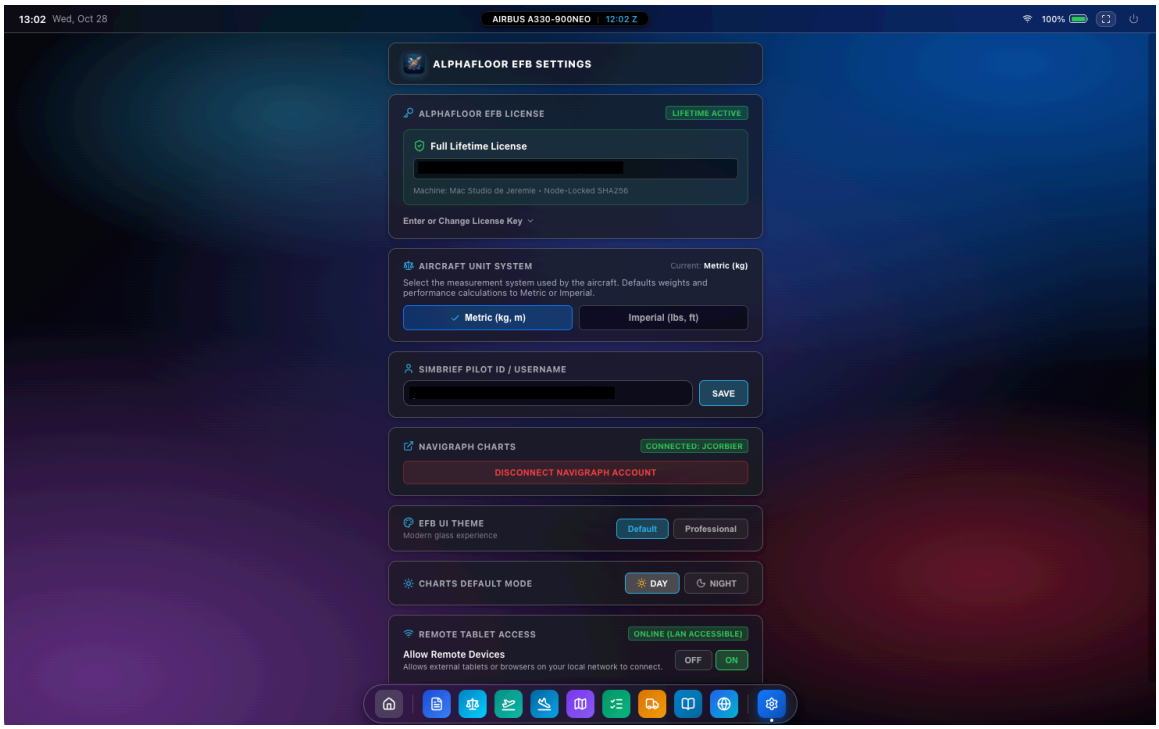


Figure 19: Settings & System Configuration

14.1 1. License Management

- **License Status**: Displays your current entitlement:
  - **Licensed (Full Lifetime Version)**
  - **Active Free Trial (X days remaining)**
- **Activate License**: Input your purchased license key (`key/XXXX`) and click **Activate**.

14.2 2. Aircraft Unit System

- **Unit System**: Select your aircraft's unit system:
  - **Metric (kg, m)**: Weights in Kilograms (kg) and Tonnes (t), barometric pressure in hPa.
  - **Imperial (lbs, ft)**: Weights in Pounds (lbs) and 1000 lbs (klbs), barometric pressure in inHg.
- This setting defines the default unit system used by the Performance and Weight & Balance applications.

14.3 3. Account Integrations

- **SimBrief**: Enter and save your SimBrief Pilot ID or username to automatically load dispatch flight plans.
- **Navigraph**: View your Navigraph Charts connection status, authenticate, or sign out.

14.4 4. Web Browser Settings

- **Default Search Engine**: Choose your preferred search provider for the Web Browser address bar (**Google** or **DuckDuckGo**).

14.5 5. Charts Mode

- **Charts Mode**: Toggle between dark and bright chart background.

14.6 6. Remote Access (Wi-Fi LAN Streaming)

- **Server Status**: Indicates whether the local network server is active.
- **Local LAN Address**: Displays the exact URL to type into your external tablet or phone browser (e.g., `http://192.168.1.42:40899`).
- **Connection QR Code**: Scan the QR code with your mobile camera to open the EFB interface without typing IP addresses.
- **Port Configuration**: Change the default listening port (`40899`) if you have other software utilizing that port.

14.7 7. Synchronize EFBs

- **Synchronize EFBs**: Enable or disable synchronization between all connected EFBs.

14.8 8. Virtual Keyboard

- **Virtual Keyboard**: Enable or disable the in-cockpit on-screen virtual keyboard when selecting text or numeric inputs. When set to **OFF**, selecting an input field allows typing exclusively through your physical hardware keyboard without displaying an on-screen keyboard overlay.

15 Troubleshooting & FAQ

This section provides solutions for common setup questions, network connectivity, and troubleshooting tips.

> The **AviTab** plugin should be disabled when using AlphaFloor EFB to avoid any unexpected interactions.

15.1 Remote Tablet & Wi-Fi Streaming Issues

15.1.1 My iPad or tablet cannot open the EFB web page

- **Same Wi-Fi Network**: Ensure your mobile device and simulation computer are connected to the exact same Wi-Fi router or subnet. If your phone or tablet is connected to cellular 4G/5G or a guest Wi-Fi network with client isolation enabled, it cannot reach the PC.
- **Verify IP Address**: Simulation PCs can change IP addresses if your router reboots. Check the **Settings** screen on the cockpit tablet to confirm the current IP address.
- **Firewall Rules**: Ensure your operating system firewall allows incoming TCP connections on the configured EFB port (default `40899`).

15.2 SimBrief & Charts Issues

15.2.1 My SimBrief flight plan does not load

- Confirm that your **SimBrief Username** or **Pilot ID** is entered correctly in the EFB Settings without extra spaces.
- Ensure you have actively generated an Operational Flight Plan (OFP) on SimBrief.com before clicking Fetch.
- Verify that your simulation computer has an active internet connection.

15.2.2 Navigraph charts do not display

- Verify that your Navigraph account has an active Charts subscription.
- Go to **Settings -> Navigraph**, tap **Sign Out**, and complete the authentication process again using the QR code or device code.

15.3 Weather Issues

15.3.1 Profile view shows “Weather data temporarily unavailable”

- The profile view queries the Open-Meteo weather API to generate atmospheric soundings along your route.
- Open-Meteo enforces request rate limits. If these limits are reached or the service is temporarily unreachable, this overlay is displayed until the limit resets. Weather data is automatically cached per flight once retrieved to avoid unnecessary requests.

15.4 Licensing Issues

15.4.1 My license key shows “Invalid Key”

- Ensure there are no accidental spaces before or after the key when copy-pasting.
- License keys follow the format: `key/XXXX`.

- If you purchased recently, ensure your payment was completed successfully on Stripe.

15.4.2 Standalone file `AlphaFloorEFB.lic` is not detected

- Verify that the file is placed directly in the main plugin folder:

  ```
  X-Plane 12/Resources/plugins/AlphaFloorEFB/AlphaFloorEFB.lic
  ```

- Make sure your operating system has not appended an extra extension (such as `AlphaFloorEFB.lic.txt`).

15.5 Contact Support

If you encounter an issue not covered in this guide, our support team is available to assist:

- **Support Email**: contact@cyberpatate.fr
- **Website**: https://cyberpatate.fr/alphafloorefb/